Python Activity-Based Learning

Case Study 7: Retail Billing with Multiple Discount Rules

1. Problem Statement

Create an itemized bill for multiple products. Calculate quantity-based totals, category-specific tax, threshold discounts, and a maximum discount limit.

2. Objectives

- Store product details in lists.
- Calculate item totals, tax, and discount using loops and conditions.
- Print an itemized invoice and final payable amount.

3. Assumptions

- Each product has a name, price, quantity, and category.
- Electronics tax is 18%; groceries tax is 5%; all other categories tax is 12%.
- A 10% discount applies when the pre-tax subtotal is at least ₹2,000, but the discount cannot exceed ₹500.
- Tax is calculated on the subtotal after discount.

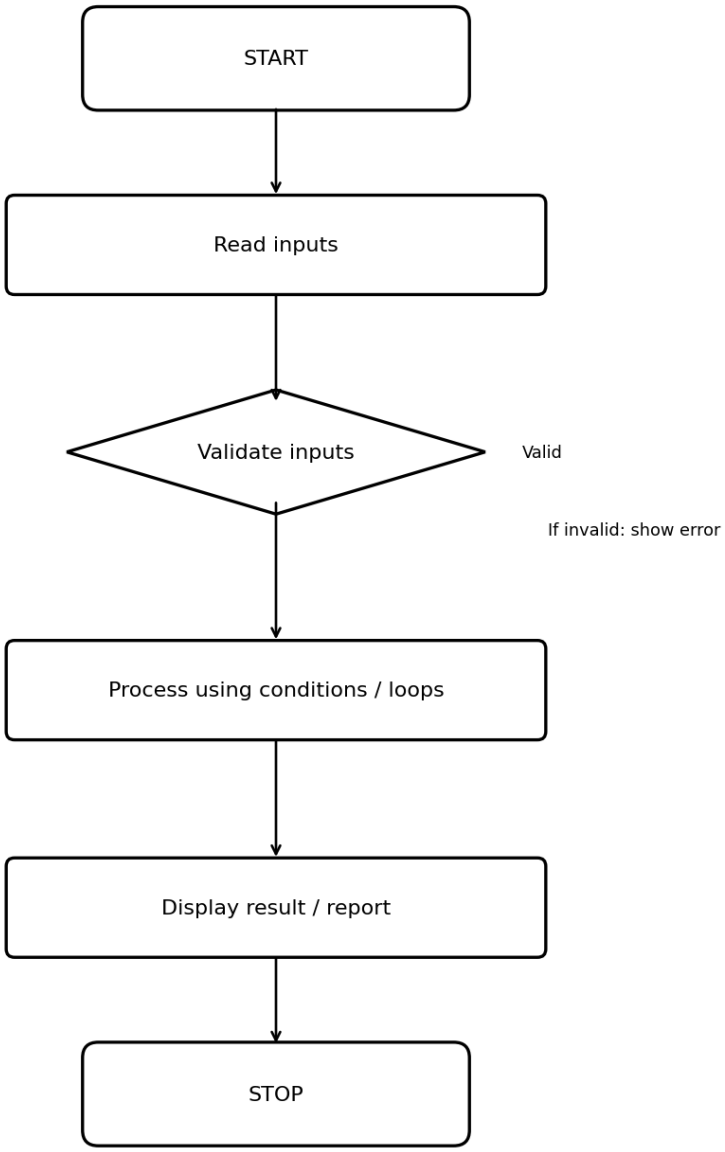
4. Algorithm

1. Read the number of products.
2. For each product, read name, price, quantity, and category.
3. Validate price and quantity.
4. Calculate line total and add it to the subtotal list.
5. Calculate discount based on the threshold and maximum limit.
6. Calculate category-wise tax for each item proportionally after discount.
7. Display each line and the final amount.

5. Flowchart

The diagram below shows the overall program flow. The validation decision represents the input checks described in the algorithm.

Flowchart: Retail Billing with Multiple Discount Rules



6. Pseudocode

```
START
READ number_of_products
subtotal = 0
CREATE empty list items
FOR each product
    READ name, price, quantity, category
    IF price < 0 OR quantity <= 0 THEN DISPLAY invalid product
    ELSE
        line_total = price * quantity
        APPEND product details and line_total to items
        subtotal = subtotal + line_total
    END IF
END FOR
IF subtotal >= 2000 THEN discount = minimum(10% of subtotal, 500)
```

```
ELSE discount = 0
FOR each item
    item_discount = discount * item_line_total / subtotal
    taxable = item_line_total - item_discount
    SELECT tax rate using category
    tax = taxable * tax_rate
    ADD tax to total_tax
END FOR
DISPLAY invoice, discount, tax, and grand total
STOP
```

7. Python Source Code

The solution uses only basic Python features from the assigned syllabus.

```
  count = int(input("Number of products: "))
  items = []
  subtotal = 0

  for i in range(count):
      print(f"Product {i + 1}")
      name = input("Name: ")
      price = float(input("Price (₹): "))
      quantity = int(input("Quantity: "))
      category = input("Category (electronics/grocery/other): ").lower()

      if price < 0 or quantity <= 0:
          print("Invalid product. Skipped.")
          continue

      line_total = price * quantity
      items.append((name, price, quantity, category, line_total))
      subtotal += line_total

  if subtotal >= 2000:
      discount = min(subtotal * 0.10, 500)
  else:
      discount = 0

  total_tax = 0
  print("\n--------- ITEMIZED INVOICE ---------")
  for name, price, quantity, category, line_total in items:
      if subtotal > 0:
          item_discount = discount * line_total / subtotal
      else:
          item_discount = 0
      taxable = line_total - item_discount

      if category == "electronics":
          rate = 0.18
      elif category == "grocery":
          rate = 0.05
      else:
          rate = 0.12

      tax = taxable * rate
      total_tax += tax
      print(f"{name}: {quantity} x ₹{price:.2f} = ₹{line_total:.2f}; "
```

```
            f"tax ₹{tax:.2f}")

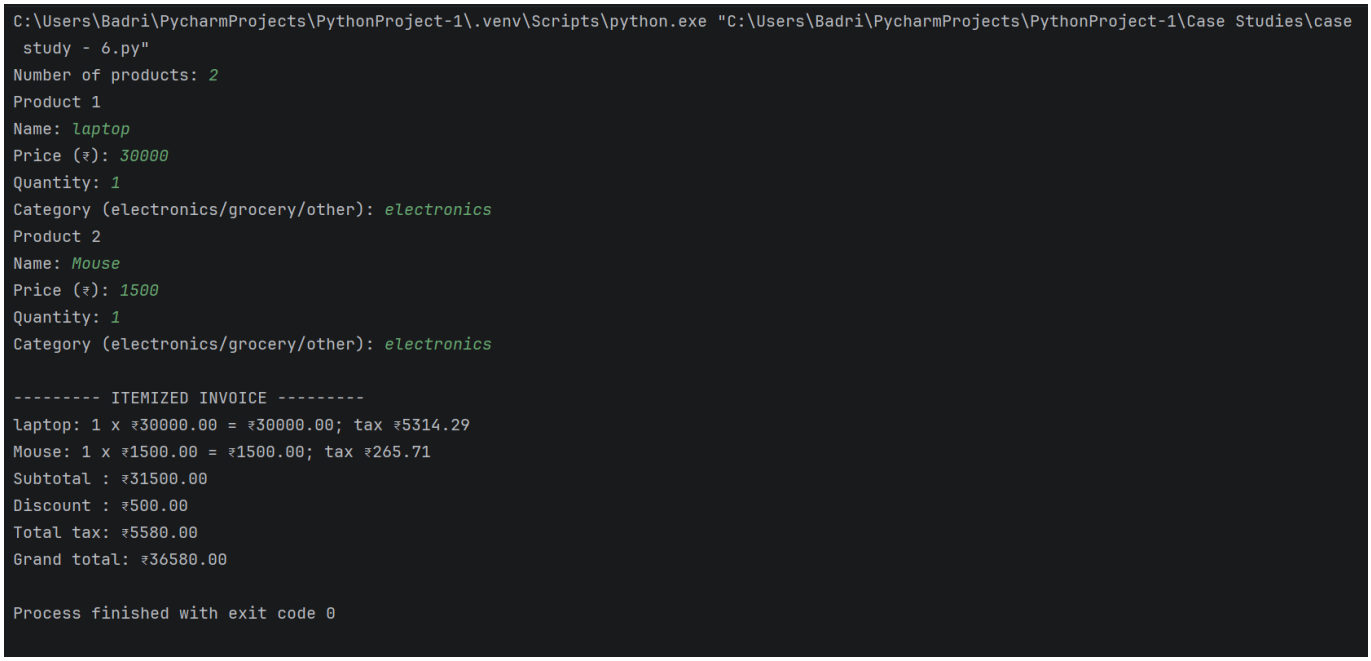
    grand_total = subtotal - discount + total_tax
    print(f"Subtotal : ₹{subtotal:.2f}")
    print(f"Discount : ₹{discount:.2f}")
    print(f"Total tax: ₹{total_tax:.2f}")
    print(f"Grand total: ₹{grand_total:.2f}")
```

8. Test Cases

Test input / condition	Expected result	Purpose
Laptop ₹30000 x1 electronics	Subtotal ₹30,000	Discount capped at ₹500; 18% tax applied to discounted amount
Rice ₹100 x5 grocery	Subtotal ₹500	No discount; 5% tax
Quantity 0	Invalid quantity	Product skipped

9. Sample Output Screenshot

Sample console output for one test run:



10. Conclusion

The program combines lists, tuples, loops, conditional tax rates, discount rules, and formatted invoice output.

Note: Assumed rates, thresholds, and sample values are for this educational exercise and can be changed if the instructor supplies different rules.